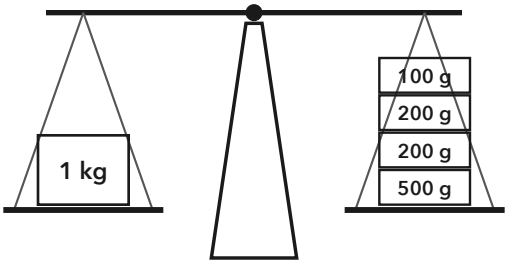
Mass and Liquid Volume

How heavy is it? How much does it hold?

Grams for light things, kilograms for heavy things

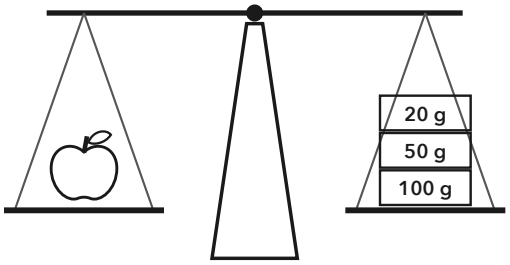
A paper clip is about 1 gram (g).



1 kilogram = 1000 grams

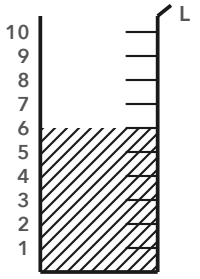
The pans are level: they weigh the same.

Read the balance: add the weights



100 + 50 + 20 = 170 g

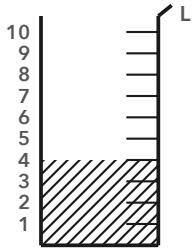
Read the beaker: count the marks



6 liters (L)

Each mark is 1 liter.

Your turn!



4 liters

For teachers

Goal: measure mass in grams and kilograms and liquid volume in liters, and solve problems with them (3.MD.2). Hold a paper clip (1 g) and a big book (about 1 kg). Say "Is it light or heavy? Grams or kilograms?"

Mass and Liquid Volume: Practice

Grams, kilograms and liters.

Part 1 Grams or kilograms? Light things are grams. Heavy things are kilograms.

1) **A grape is about 5** (grams) kilograms

2) **A big dog is about 30** grams (kilograms)

3) **A paper clip is about 1** (grams) kilograms

4) **A bike is about 12** grams (kilograms)

5) **A strawberry is about 15** (grams) kilograms

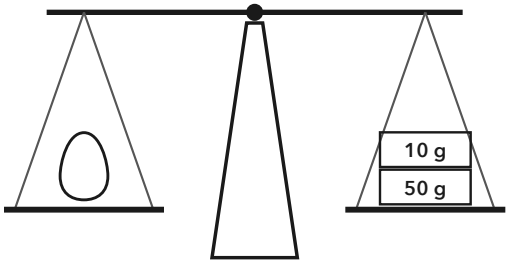
6) **A bag of rice is about 2** grams (kilograms)

7) **A pencil is about 6** (grams) kilograms

8) **A watermelon is about 5** grams (kilograms)

Part 2 Read the balance Add the weights on the other pan.

9)

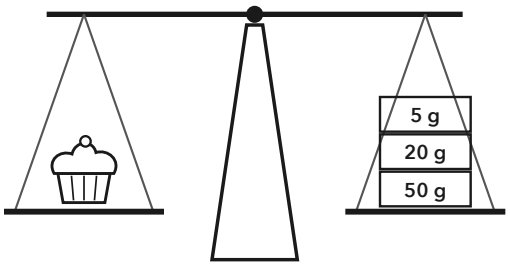


50 + 10 = [60] **g**

Mass and Liquid Volume: Practice (continued)

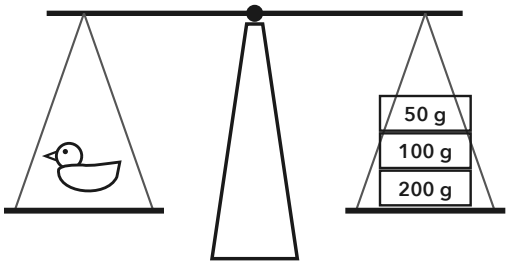
Part 2 Read the balance (continued)

10)



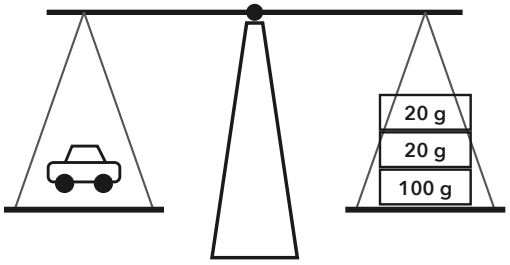
50 + 20 + 5 = 75 g

11)



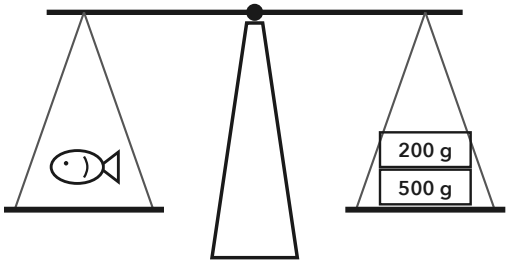
200 + 100 + 50 = 350 g

12)



100 + 20 + 20 = 140 g

13)

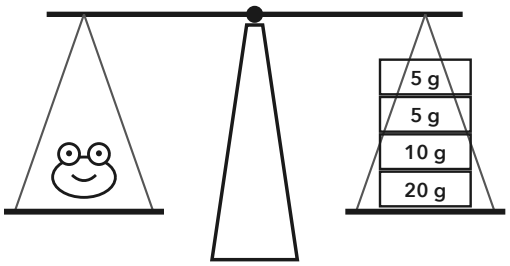


500 + 200 = 700 g

Mass and Liquid Volume: Practice (continued)

Part 2 Read the balance (continued)

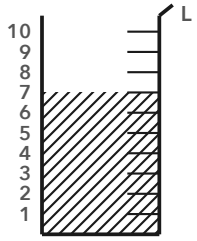
14)



20 + 10 + 5 + 5 = 40 g

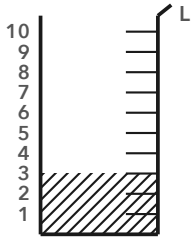
Part 3 Read the beaker Find the top of the water. Count the marks.

15)



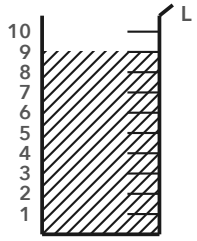
7 liters

16)



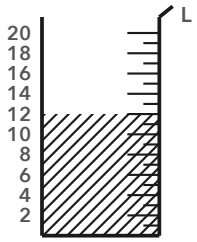
3 liters

17)



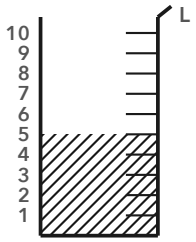
9 liters

18)



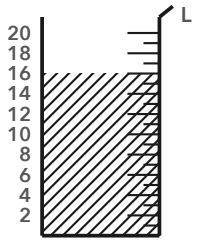
12 liters

19)



5 liters

20)

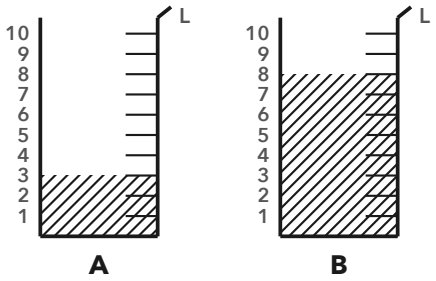


16 liters

Mass and Liquid Volume: Practice (continued)

Part 4 How much more? Take away the smaller amount.

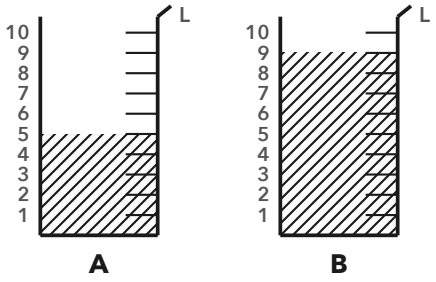
21)



How much more is in B?

8 L − 3 L = 5 L

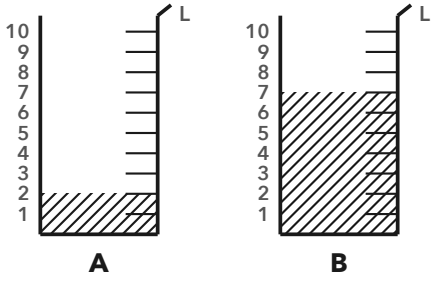
22)



How much more is in B?

9 L − 5 L = 4 L

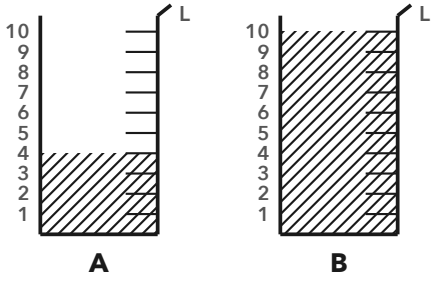
23)



How much more is in B?

7 L − 2 L = 5 L

24)



How much more is in B?

10 L − 4 L = 6 L

Mass and Liquid Volume: Practice (continued)

Part 5 Add and take away grams No help this time.

25) 350 g + 240 g = 590 g

26) 800 g − 350 g = 450 g

27) 125 g + 475 g = 600 g

28) 1000 g − 400 g = 600 g

29) 460 g + 270 g = 730 g

30) 900 g − 625 g = 275 g

31) 3 kg + 6 kg = 9 kg

32) 15 kg − 8 kg = 7 kg

Part 6 Liters No help this time.

33) 4 bottles of 2 L = 8 L

34) 3 jugs of 5 L = 15 L

35) 12 L in 3 equal jugs = 4 L each

36) 7 L + 8 L = 15 L

37) 20 L − 6 L = 14 L

38) 6 buckets of 4 L = 24 L

Part 7 Remember? Problems from earlier packets.

39) 6 × 8 = 48 so 6 × 4 = 24

40) 4 m by 7 m: area = 28 sq m

41) 25 minutes after 9:50 = 10:15

42) Round 638 to the nearest ten: 640

43) 81 ÷ 9 = 9

44) 306 + 495 = 801

Mass and Liquid Volume: Practice (continued)

Part 8 Word problems Draw a bar if it helps.

45) A bag of apples weighs 3 kg. How much do 4 bags weigh?

Number sentence $4 \times 3 = 12$

Answer 12 kg

Draw or show your work

46) A pitcher holds 2 liters. How many pitchers fill an 18-liter tub?

Number sentence $18 \div 2 = 9$

Answer 9 pitchers

Draw or show your work

47) Mia's backpack weighs 4 kg. Her lunch adds 1 kg and a book adds 2 kg. What is the total?

Number sentence $4 + 1 + 2 = 7$

Answer 7 kg

Draw or show your work

48) A fish tank holds 40 liters. It has 23 liters in it now. How much more water will fill it?

Number sentence $23 + 17 = 40$

Answer 17 liters

Draw or show your work

49) Challenge: A box holds 6 bags of flour. Each bag is 2 kg. The empty box is 1 kg. How heavy is the full box?

Number sentence $6 \times 2 = 12,\ 12 + 1 = 13$

Answer 13 kg

Draw or show your work

Check-In: Mass and Liquid Volume

Show what you know. No help this time.

Mastery: 5 of 6 (80%)

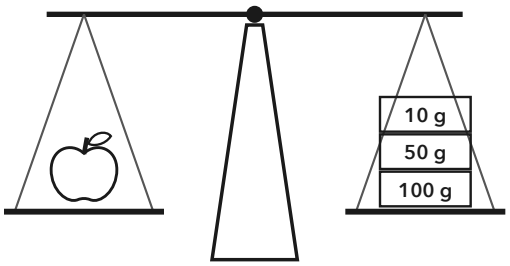
Part 1 Grams or kilograms?

1) **A coin is about 3** (grams) kilograms

2) **An eraser is about 20** (grams) kilograms

Part 2 Read the balance

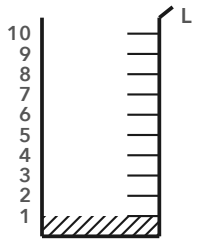
3)



100 + 50 + 10 = 160 g

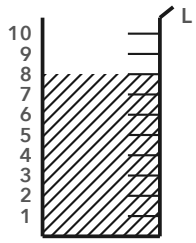
Part 3 Read the beaker

4)



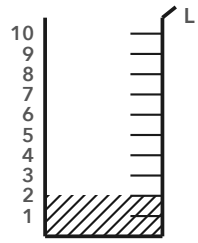
1 liters

5)



8 liters

6)



2 liters